Does Entropy Help Detect IoT Faults? A Comparative Study with Generative AI

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Abstract. A faulty sensor may begin copying its neighbors while producing apparently plausible readings. Detecting such changes requires examining both relationships between sensors and patterns within individual streams. We investigate spatial and temporal entropy as complementary measurements for IoT fault detection through a GPU implementation that compares conventional statistics with entropy features under historical resampling and generative diffusion references. Experiments cover 360 fault realizations across 20 simulations and recording blocks, using one synthetic dataset and two real datasets with injected faults. Temporal entropy and autocorrelation capture changes the spatial covariance can ignore. On Intel data, temporal entropy with resampling achieves recall 0.389 versus 0.000 for conventional temporal statistics, with respective background alarm rates of 0.103 and 0.000. Missing data analysis helps explain observed performance differences. The findings set complementary measurement capabilities and provide practical guidance to evaluating entropy and generative models in sensor networks, with the benefits depending on fault mechanism, measurement availability and calibration.

Keywords: IoT · spatial entropy · permutation entropy · sample entropy · diffusion forecasting · fault localization

1 Introduction

Sensors in digital twins and data spaces need evidence about changes in both collective organization and individual dynamics. Copying streams can reduce independent spatial variation, replay can alter the temporal organization without creating a large marginal outlier. A legitimate operating transition can also change both. An alert should therefore expose measurements and their provenance, rather than assert an unverified physical cause.

This study compares information captured by spatial and temporal entropy with conventional measurements, separating the features from its normal reference model. There is a motivating connection to prior work, that entropy trajectories describe evolving homogeneous clusting under different network topologies [9] and collective flocking organization [10]. We transfer attention to evolving organization and interaction structure. The segregation study uses degree/homophily distributions, the flocking work develops a thermodynamic description of its simulation. Here spatial entropy uses normalized correlation eigenvalues, permutation entropy uses ordinal patterns, and sample entropy uses matching templates. These definitions differ. We infer neither thermodynamic quantities nor a second law for sensor recordings.

The earlier version of this project evaluated spatial correlation spectrum entropy and established no consistent improvement over its synchronization comparators. That finding motivates, but does not predetermine, this extension. A changing spatial entropy trace is not itself a within sensor temporal entropy estimator. Nor does an unsuccessful reference model establish a limitation of all entropy features. The original artifacts remain identifiable, new comparisons, support corrections and diagnostics have separate manifests.

We address six practical and theoretical questions. **RQ1**: does spatial entropy improve detection or localization over non-entropic spatial statistics? **RQ2**: do temporal entropies reveal changes missed by spatial covariance summaries? **RQ3**: do they improve over conventional temporal statistics? **RQ4**: does combining spatial and temporal evidence improve detection or localization? **RQ5**: does a generative reference improve over joint block bootstrap? **RQ6**: how do support, missingness, window length, calibration and mechanism affect these answers?

Our contribution is a reproducible comparison, elementary theoretical characterization of the information retained by its measurements, and an inspectable monitoring artifact. We introduce neither an entropy definition nor a general purpose diffusion architecture. Conventional features augmented with entropy are compared directly with conventional features alone; entropy is not required to win or lose.

2 Related Work

Effective rank measures spectral dimensionality through entropy of normalized singular values [14], these equal eigenvalues for a PSD correlation matrix. Graph entropy anomaly monitoring also predates this study [3], although a graph Laplacian spectrum differs from the correlation spectrum used here. Spatial meaning comes from physical/proximity groups, not from the ordering of sensor columns.

Bandt and Pompe's permutation entropy summarizes neighboring ordinal patterns [1]. Richman and Moorman's sample entropy describes template-extension regularity [13]. Their probability objects, support requirements and amplitude sensitivity differ. Costa et al's multiscale sample entropy [6] explicitly warns against identifying irregularity with healthy complexity. We separately label coarse-grained permutation and sample entropy and make no physiological health inference.

TimeGrad [12], CSDI [16] and DiffSTG [17] establish diffusion forecasting, conditional imputation and graph-conditioned generation. DiffAD [18], ImDiffusion [4] and ICDiffAD [19] already apply diffusion to anomaly detection. Our generator uses a forecasting boundary: it receives no measured target value. GDN [7] supplies a graph-prediction comparator, with adaptation and information access disclosed below. Data dependent conformal work [5] motivates explicit calibration assumptions, we do not claim a new distribution free guarantee for an indefinite sensor stream.

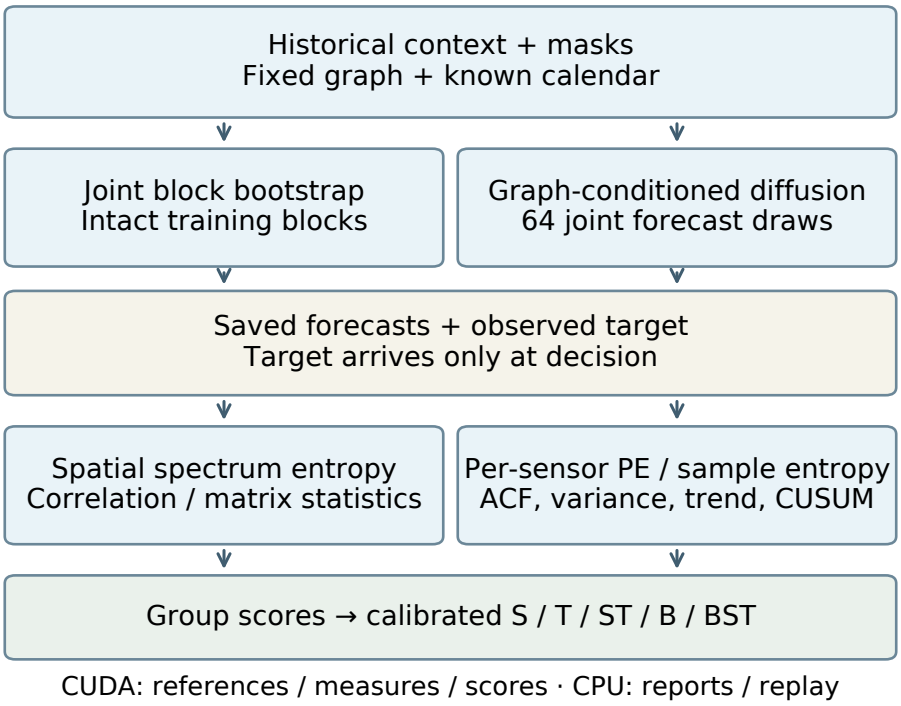


Fig. 1. Feature family and normal reference are separate experimental factors. Both generators issue a joint trajectory before the target, support is applied after generation. Calibration includes the complete frozen scan family.

3 Measurements, References and Decisions

3.1 Groups, channels and spatial measurements

We examine sensor groups to measure how their readings change together. Groups contain $m \in \{6, 12, 24\}$ sensors around 24 centers selected at regular intervals in node index order. Sensors are selected by graph distance, measured in hops, with geographic distance resolving ties. If a connected component contains too few sensors, additional nodes are selected by node order. Duplicate groups of the same size are removed. The graphs describe coupling in the synthetic system, coordinate proximity in Intel data, and the supplied road distances in PEMS. Each measurement channel is analyzed separately. The main analysis uses windows of $W \in \{48, 96\}$ observations and compares measurements $d = W/4$ steps apart. The original $W = 24$ analysis is retained separately.

For each group and channel, we use only timestamps at which every sensor has an observation. The number of usable rows must satisfy $n \geq \max(2m, \lceil .8W \rceil)$. If any sensor has sample variance at most 10^{-8} in units scaled using the training data, the correlation calculation is marked undefined and flagged for data quality. Otherwise, we center each sensor column and scale it to unit sample variance, producing the matrix Z. We then

calculate

$$\mathbf{R}_{g,t} = Z^\top Z/(n-1), \tag{1}$$

$$\widetilde{\mathbf{R}}_{g,t} = (1-\lambda)\mathbf{R}_{g,t} + \lambda I, \tag{2}$$

$$p_k = \lambda_k(\widetilde{\mathbf{R}}_{g,t})/\mathrm{tr}(\widetilde{\mathbf{R}}_{g,t}), \tag{3}$$

$$H_S(g,t) = -\sum_k p_k \log p_k / \log m, \tag{4}$$

$$\Delta H_S(g,t) = H_S(g,t) - H_S(g,t-d). \tag{5}$$

Here $\mathbf{R}_{g,t}$ is the correlation matrix for group g at time t. The identity contribution, with $\lambda = .05$, stabilizes the matrix, and p_k gives the proportion of its trace associated with eigenvalue k. Entropy is lower when variation is concentrated in a few shared patterns and higher when it is distributed more evenly across patterns. The signed change ΔH_S records both increases and decreases. This is *local correlation-spectrum entropy*, with spatial meaning provided by the sensor groups. We use $0 \log 0 = 0$. The symmetric eigensolver rejects negative eigenvalues beyond the tolerance value of $64m$ times machine precision. TF32 is disabled when computing $Z^\top Z$.

Conventional spatial measurements include mean signed correlation C, mean absolute correlation, standardized pairwise disagreement D, the leading eigenvalue concentration $\lambda_{\max}(\mathbf{R})/m$, and their changes over the same interval. For the standardized columns, $D = 2(n-1)(1-C)/n$. Thus, when the number of usable rows is fixed, D contains the same information as C. We also compare the complete correlation matrix $\mathbf{R}$ and its change $\Delta\mathbf{R}$ against the reference trajectories using standardized Frobenius discrepancies. These comparisons capture changes across all sensor pairs. Entropy compresses this matrix into one value and cannot add information to it, although the summary can behave differently during estimation and calibration.

3.2 Temporal entropy and data requirements

We calculate each sensor's entropy before aggregation. For sensor i, permutation entropy measures the diversity of orderings in $v_{i,s} = (x_{i,s+j\tau})_{j=0}^{q-1}$. Window frequencies $p_{i,t}(\pi)$ give

$$H_{\mathrm{PE}}(i,t) = -\sum_\pi p_{i,t}(\pi) \log p_{i,t}(\pi) / \log(q!). \tag{6}$$

Development compares $(q,\tau) = (3,1), (3,2), (4,1)$, starting from $(3,1)$. Missing observations invalidate affected templates without closing time gaps. Equal values are ordered by timestamp. Both windows require $\max(30, 5q!)$ valid templates. Flatlines and tied fractions above .5 are excluded from scoring. We record tied fractions and test excluding tied templates during development. For untied values, strictly monotone amplitude transformations preserve entropy. Higher entropy need not indicate healthier behavior.

Sample entropy measures how often similar sequences remain similar after another observation. Using complete $(r+1)$-templates with $r = 2$, we form pairs $\mathcal{P}$ with starts $a < b$ and $b - a > 2$, excluding self matches and the Theiler neighborhood. Let D_r denote

the maximum absolute difference across the first r coordinates, with D_{r+1} including the next observation. Then

$$B_r = \sum_{(a,b)\in\mathcal{P}} \mathbf{1}\{D_r(a,b) \le \delta\}, \tag{7}$$

$$A_{r+1} = \sum_{(a,b)\in\mathcal{P}} \mathbf{1}\{D_{r+1}(a,b) \le \delta\}, \tag{8}$$

$$H_{\mathrm{SE}}(i,t) = -\log(A_{r+1}/B_r). \tag{9}$$

Both counts use identical pairs. Tolerance candidates .1, .2 and .3 multiply the fixed training scale. Development maximizes the fraction of finite estimates, with predefined tie preferences. $B_r = 0$ gives an undefined estimate, while $A_{r+1} = 0 < B_r$ gives a censored infinite value. Scoring requires 30 templates, $B_r \ge 20$, $A_{r+1} > 0$, nonconstant data and valid comparison windows. No pseudocounts or finite substitutes are used. Match, pair and template counts remain visible alongside censoring.

Both entropies provide a level and lagged change. Multiscale diagnostics average nonoverlapping bins of 1, 2 or 4 observations from the window's left boundary. Bins with missing observations are invalid. Incomplete final bins are discarded. Template requirements and forecast horizon remain unchanged. Conventional comparators use identical histories. A separately frozen exploratory comparison evaluates scales 1 and 2 on $W = 96$, retaining $d = 24$ original time steps, 17 episodes and references. Scale 2 floors are refitted using the original four development issuances, and each family is recalibrated. Other settings remain fixed. Permutation, autocorrelation and Theiler lags follow the coarse grid, doubling their physical delays. Spatial measurements use the original grid.

3.3 Conventional temporal measurements and feature families

Temporal baselines measure persistence through autocorrelation (ACF) at lags 1, 2 and 4 and changes in this vector. Variability is measured by first difference variance and local sample variance, drift by a least squares trend, and mean departures by predictive deviation and a two-sided CUSUM. Lag pairs preserve actual time positions. Each statistic requires at least $\max(12, \lceil W/2 \rceil)$ observations or valid pairs. CUSUM resets at each window, uses allowance .5 in training-scaled units, pauses on missing rows, and reports its final state. Reference scoring expresses measurements as departures from expected behavior.

Additional detectors use raw predictive residuals and the retained GDN adaptation. Raw residuals average absolute pointwise deviations divided by (generated MAD plus .05), retaining the original spatial quality restriction. GDN uses 32-dimensional embeddings, learned top-15 neighborhoods, graph attention and a shared two-layer head for multiple channels without batch normalization. Errors standardized on development data are averaged within four trailing one step predictions before taking the group maximum. GDN receives causal observed target history, giving it more recent information than the long-lead references. Its group support and participation mapping are adaptations, so published GDN benchmark accuracy is not claimed. Feature families comprise spatial entropy (S), permutation entropy (PE), sample entropy (SE), both temporal entropies

(T), and all entropies (ST). Conventional families are spatial (SB), temporal (TB), and combined (B). BST adds all entropies to B. Families use the maximum component score. Unsupported components are skipped, with abstention if none qualify. Level/change pairs require both endpoints. Comparing B with BST tests whether entropy improves conventional detectors.

3.4 Joint bootstrap and generative references

Both references generate possible future readings jointly across sensors, using 48 observations to predict 120 steps. Inputs include calendar cycles, coordinates and graph connections, but exclude future readings. Forecasts for decision t are issued at $t - 120$. Current and earlier windows use steps 25 to 120 and 1 to 96 for $W = 96$, or 73 to 120 and 61 to 108 for $W = 48$. Both windows share each generated trajectory. This means forecasting 240 minutes ahead for Intel and 600 for PEMS.

Bootstrap resamples intact blocks from the 64 training donors most similar in preceding readings and calendar position. The diffusion model is inspired by DiffSTG/CSDI, using width 24, three graph-temporal blocks, historical readings and missingness indicators, and fixed graph/calendar encodings. Training adds Gaussian noise ϵ to trajectories Y and learns to predict that noise

$$Y_s = \sqrt{\bar{\alpha}_s} Y + \sqrt{1 - \bar{\alpha}_s}\epsilon, \tag{10}$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\left[\frac{\|M_{\text{loss}} \odot (\epsilon - \epsilon_\theta(Y_s, s, C, G))\|_F^2}{\sum M_{\text{loss}}}\right]. \tag{11}$$

Here s indexes diffusion steps and $\bar{\alpha}_s$ controls retained signal. The network ϵ_θ uses context C and graph G to predict noise. The mask M_{loss} restricts training error to observed targets. Training uses a cosine schedule [11], 100 diffusion steps, Adam at 10^{-3}, batches of 8, up to 2400 updates and development early stopping. Generation starts from noise using 20 step DDIM [15], clipping denoised states to $[-8, 8]$ training units. Seeds 17/29/43 are retained without architecture tuning on inspected tests. Each method generates 64 trajectories. Observed missingness and additional dropout are applied after generation. Bootstrap donors retain further gaps. Even for identical evaluation windows, usable row and template counts can therefore differ.

3.5 Scores, group aggregation and calibration

For feature a, set $u_a = (f_a, f_a - f_{a,t-d})$. Componentwise generated medians $c_{a,t,j}$ and MADs give

$$z_{a,t,j} = \frac{u_{a,t,j} - c_{a,t,j}}{1.4826 \ \text{median}_b \ |u^{(b)}_{a,t,j} - c_{a,t,j}| + \epsilon_j}, \tag{12}$$

$$A_a(t) = \max_j |z_{a,t,j}|. \tag{13}$$

Floors for level/change features are $\max(10^{-4}, .05\,\text{IQR}_{dev})$. Legacy matrix distance and raw residual comparators retain .01 and .05 regularization floors, respectively. Reference

support counts are retained, the temporal rule requires at least 52 finite draws, while the retained spatial summary implementation uses 51 of 64. MAD scaling does not imply Gaussian tails. Actual Δf remains separate from $\Delta f - c_{\Delta f}$.

For each temporal group, average its largest $k = \lceil \eta m \rceil$ valid sensor scores, requiring at least $\lceil m/2 \rceil$ valid members. Fractions $\eta \in \{.25, .5\}$ and localization budgets in $\{6, 12, 24\}$ are selected jointly from four development copy/noise/drift/delay episodes. Mean localization IoU is pooled over T, TB, ST, B and BST and both references. All families use the same selected fraction and budget. Max combined families receive their own calibration, no uncalibrated union of separately thresholded alerts is used.

With frozen family $\mathcal{F}$, let $M_t = \max_{a \in \mathcal{F}} A_a(t)$. Calibration units are separated by 168 rows (48 context plus 120 target), or are separate synthetic simulations. For n_c calibration maxima,

$$p_t = \frac{1 + \sum_{b=1}^{n_c} \mathbf{1}\{M_b \geq M_t\}}{n_c + 1}. \tag{14}$$

An entirely abstained unit has $M = -\infty$ and cannot alert. Primary $\alpha = .10$ is fixed before testing. Temporal dependence and distribution shift can invalidate exchangeability despite gaps; this is a per-issuance statement, not a stream-wide guarantee. Neither $1 - p$ nor a model deviation is a physical fault probability. These test controls evaluate background rates; separate retrospective rate caps are clearly descriptive, not the prospective threshold validation.

3.6 Localization and availability

Group participation weights each group's score by inverse size and normalizes over eligible incident groups. Direct temporal ranking uses sensor scores, the combined rule averages their normalized rank percentiles. S/SB use participation, T/PE/SE/TB direct ranking, and ST/B/BST the combined rule. All three alternatives are also reported. The budget is selected on development data, never from test truth. Candidate group best overlap IoU is an oracle diagnostic, not a deployable method. Participation is not causal attribution.

Operational analysis includes every scheduled unit and injected event, including abstentions and misses. A separately calibrated common support analysis uses group/window/channel units valid for S, PE, SE, SB and TB within each reference. Reference comparisons also report their differing model support counts, common feature support is not claimed to eliminate every reference support difference. Conditional localization is reported only alongside unconditional scores, where misses contribute zero. Quality flags, signed agreement and onset alignment are descriptive and do not create additional uncalibrated detections.

4 Theoretical and Numerical Characterization

Proposition 1 (Bounds and shrinkage). *For a PSD correlation matrix,* $0 \leq H_S \leq 1$. *At* $\lambda = 0$*, identity has entropy one and rank one entropy zero. With positive shrinkage the lower bound is* $-[a \log a + (m-1)b \log b]/\log m$*, where* $a = 1 - \lambda + \lambda/m$ *and* $b = \lambda/m$.

Proof. The eigenvalues normalized by trace form a probability vector. Shannon entropy is nonnegative and maximized by uniform probabilities. Shrinkage maps its simplex into the convex hull of permutations of $(a, b, \ldots, b)$; concavity makes these extreme points minimizers.

For equicorrelation $\mathbf{R}(\rho) = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^\top$,

$$\frac{dH_S}{d\rho} = \frac{m-1}{m\log m}\log\frac{1-\rho}{1+(m-1)\rho} \le 0, \quad 0 \le \rho < 1. \tag{15}$$

The eigenvalues are $1+(m-1)\rho$ and $1-\rho$ with multiplicity $m-1$. Substitution proves the derivative; shrinkage replaces ρ by $(1-\lambda)\rho$ and multiplies by $1-\lambda$. The displayed nonpositive sign is restricted to nonnegative ρ, not the whole PSD interval.

Entropy need not be determined by average correlation. For $m = 4$, compare equicorrelation .2 with two independent two-sensor blocks of correlation .6. Both have $C = C_{abs} = .2$ and leading concentration .4, but unshrunk entropies are .960964 and .860964. Their full matrices distinguish them. This is an analytic example, not empirical evidence of detector superiority.

Proposition 2 (Order invariance on an aligned complete window). *For a common time-row permutation P, the correlation matrix and its correlation-spectrum entropy are unchanged.*

Proof. Column centering and sample scaling commute with row permutation, and $(PZ)^\top(PZ)/(n-1) = Z^\top P^\top PZ/(n-1) = Z^\top Z/(n-1)$.

Order-sensitive temporal measurements can change. This elementary property applied to measurement design is not a new theorem. It does not imply invariance of every overlapping rolling window or the whole entropy-change trace. Offsets, nonzero column gains and sign conjugation also leave the correlation spectrum unchanged; temporal ordinal entropy can likewise miss amplitude changes.

Proposition 3 (Conditional rank calibration). *Conditional on a fixed scoring rule, if calibration and next-null maxima are exchangeable, (14) is super-uniform. Its smallest attainable value is* $1/(n_c+1)$.

Proof. Among $n_c + 1$ exchangeable scores the test score's upper rank is uniform in the absence of ties; including all tied calibration scores makes it conservative. The trained rule must be fixed before these units.

The premise is not established merely by chronological partitioning. Controlled null simulations, finite-difference derivatives, explicit CPU pair counts, CPU/GPU agreement and leakage/event-matching checks accompany the proof.

5 Experimental Design and Evaluation

We test whether the entropy estimation complements conventional measurements and if generative references improve fault detection. We use one synthetic dataset and two real

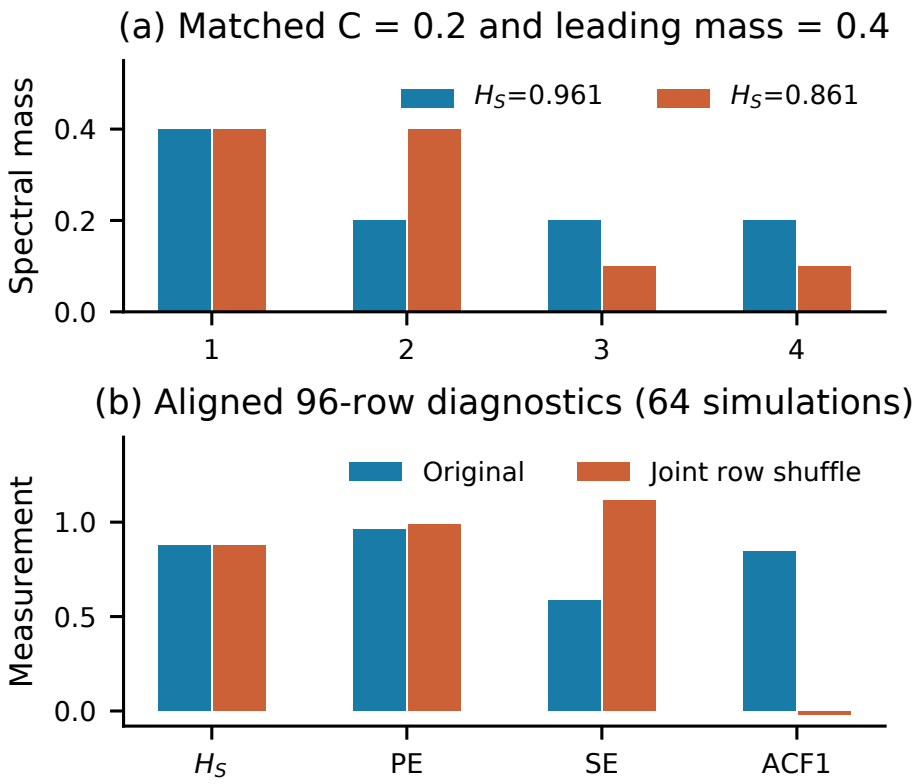


Fig. 2. Mathematical illustrations and aligned diagnostics, not primary fault performance. Equal average correlation and leading concentration can coexist with different spectral entropy. A joint time permutation preserves the evaluated matrix while temporal measurements can change.

sensor datasets with injected faults. Synthetic networks provide controlled dynamics, while real recordings introduce practical variability and missing data. Three synthetic network sizes give five configurations, with 360 fault realizations across 20 background blocks and three model seeds. The protocol was fixed before full runs. Earlier work inspected the real backgrounds, making those comparisons exploratory.

Synthetic data. Random geometric graphs contain 64, 128 or 256 sensors. Their readings evolve according to $x_{t+1} = (.82I - .25L/\max_i \deg_i)x_t + Bf_t + .12\epsilon_t$, where L is the graph Laplacian, Bf_t represents three shared Ornstein Uhlenbeck drivers with spatial loadings, and ϵ_t supplies random noise. Calendar signals add periodic variation. The transition matrix has spectral radius .82, and 256 initial steps are discarded. Training data and scaling remain fixed. Independent seeds generate four development simulations of 312 observations, 32 calibration simulations of 256 observations, and four test simulations of 312 observations per size. Windows remain within individual simulations. Display timestamps represent simulation phase.

Intel Lab. This dataset [2] contains readings from 54 sensors approximately every 31 seconds. Measurements include temperature, humidity, light and voltage. We analyze temperature and humidity, retaining the other channels for auxiliary analysis. Each 2 minute interval contributes its last available reading at the interval endpoint, preserving causal timing and missingness. Groups follow sensor coordinates. A fixed coverage rule excludes data from March 24, 2004 onward, leaving 17,970 observations on the time grid. Chronological training, development, calibration and test partitions contain 60/15/10/15% of these observations.

PEMS-BAY. The DCRNN release [8] records traffic speed in mph from 325 road sensors every five minutes during January to June 2017. Preprocessing inserts 12 missing timestamps, masks 521 zero readings under the release convention, and retains published road distances. The resulting 52,128 timestamps use the same chronological partition proportions as Intel. Records identifying individual values imputed upstream are unavailable. Although selected blocks excluded the earlier primary experiments, a broader diagnostic had inspected them. Both real datasets therefore provide previously examined backgrounds with injected faults. Actual physical faults are unverified.

Preprocessing and calibration. Measurements are centered using training medians and scaled by IQR/1.349, with documented fallbacks. Missing neural inputs use zero plus a mask, while statistical estimators retain missing values. Missing values are never filled from neighbors. Windows stay within partitions. Models and groups remain fixed. Thresholds are fitted without test labels, using the previously examined real calibration partitions. There are 32 synthetic, 10 Intel and 31 PEMS forecast issuances for calibration. Their minimum rank p-values are .0303, .0909 and .03125. Intel's ten calibration cases therefore permit only coarse threshold choices near nominal .10.

Fault scenarios. Twelve mechanisms cover copying, common interference, decoupling, independent noise, delay, offset/gain drift, flatline, dropout, anticorrelated copying, mixed collapse/noise, covariance changes preserving average correlation, and temporal rearrangements preserving marginal values. Combining severities .6/1 with durations 12/48/96 gives 72 fault realizations per configuration. Affected groups contain 6 or 12 sensors and may be connected or partly disconnected. Synthetic decoupling changes the state transition while sharing randomness with its clean counterpart. Real decoupling modifies recordings without changing the physical system. Faults begin at observation 192. Decisions occur at 167,179,. . .,311, placing the first possible detection at 203, eleven samples after onset. Forecast inputs precede every fault. Unmodified backgrounds and legitimate synchronized transitions provide controls. Faults constitute 90% of these constructed trials, so precision reflects this benchmark's event mix rather than deployment prevalence.

Alert evaluation. Consecutive flagged decisions form one alert. Hungarian matching pairs alerts and events one to one, crediting only alerts beginning during an event. Event boundaries are not extended, and point labels are not adjusted after detection. Unmatched early, late and control alerts enter benchmark precision. We report alarms on unlabelled real backgrounds as background alerts. Event precision-recall area uses the upper precision over recalls because changing thresholds can merge alerts and make recall nonmonotonically. Detection delay includes the decision schedule. Computation time is measured as a different source.

Uncertainty and comparison controls. Paired percentile intervals resample complete simulations or recording blocks, keeping their repeated injections and three model seeds together. Overlapping windows do not count as independent repetitions. Four blocks per configuration limit precision, and adjacent recording blocks may remain dependent. Intervals are exploratory, without multiple comparison adjustment or a planned equivalence test. Results are separated by mechanism, severity, duration, measured change direction and data availability. All feature comparisons fix the physical or proximity graph. Level/change pairs require valid endpoints, and matrix comparisons include both $\mathbf{R}$ and $\Delta\mathbf{R}$. This keeps scoring rules consistent across entropy and conventional features.

6 Results

6.1 Fault detection, localization and added value

Table 1 and Fig. 3 distinguish detecting faults, measured by recall, from identifying affected sensors, measured by IoU. Background rates describe the alarm burden on data without injected faults. For spatial measurements (RQ1), detection gains often accompany more background alarms. Synthetic diffusion S minus SB gives recall -0.099 $[-0.174, -0.034]$ and IoU -0.010 $[-0.017, -0.003]$. Intel diffusion S detects fewer faults than SB (0.213 versus 0.481), with fewer background alarms (0.218 versus 0.372). PEMS bootstrap S detects more faults (0.528 versus 0.005), but its background rate is 0.301 versus 0.000. Capping background rates retrospectively at .10 reverses this ordering, with recalls 0.069 versus 0.264 and rates 0.026 versus 0.064. This illustrates threshold sensitivity, without validating future alarm control.

Temporal entropy (RQ3) helps in some settings. Synthetic bootstrap T/TB recall is 0.401/0.532, a difference of -0.131 $[-0.313, +0.054]$, with IoU 0.093/0.239. Intel bootstrap T exceeds TB by $+0.389$ $[+0.264, +0.523]$ recall, at background rates 0.103 versus 0.000. Sample entropy has higher Intel recall and sensor overlap than permutation entropy, with recall 0.380 versus 0.310 and IoU 0.073 versus 0.022. Diffusion reduces Intel T recall to 0.060 while increasing background alarms to 0.288. PEMS temporal comparisons remain less specific.

Combining features (RQ4) does not automatically preserve their individual strengths. Synthetic bootstrap ST improves recall over S by $+0.123$ $[+0.028, +0.230]$, but over T by only $+0.008$ $[-0.085, +0.127]$. Their localization rules differ. Additional comparisons hold these rules fixed. Adding entropy to conventional B increases synthetic bootstrap recall by $+0.003$ $[+0.000, +0.008]$ and IoU by $+0.008$ $[+0.000, +0.017]$, using identical localization rules. Recall remains unchanged in the other five dataset/reference combinations. Conventional components usually dominate the maximum score, and each family has its own calibrated threshold. For Intel bootstrap, T detects faults while B and BST detect none at the selected threshold. This does not establish feature equivalence.

Additional detectors offer different tradeoffs. GDN synthetic/Intel/PEMS recall is 0.514/0.139/0.157, with background rates 0.135/0.436/0.038. PEMS bootstrap residuals achieve recall 0.264 at background rate 0.077. Their information access and support differ. Localization remains difficult. For detected faults, bootstrap T IoU is 0.231/0.195/0.038 across the three datasets, falling to 0.093/0.076/0.015 when missed faults count as zero.

The best candidate groups, selected using fault labels, average IoU 0.284/0.413/0.325. This measures group coverage, not a ceiling on sensor rankings.

6.2 Complementary responses to different fault mechanisms

RQ2 shows how spatial and temporal measurements complement each other. In 64 aligned simulations, rearranging time points identically across sensors preserves their correlation matrix to within 1.3×10^{-15} (rounded upward). Mean PE rises from 0.964 to 0.989 and SE from 0.586 to 1.115, while ACF1 falls from 0.845 to -0.016. Spatial entropy remains unchanged. Copying lowers spatial entropy from 0.877 to 0.134, while PE changes from 0.964 to 0.965. Additional diagnostics examine offsets, gains, sign changes, quantization, replay, periodic interference, irregularity, legitimate transitions and dropout. These diagnostics isolate responses separately from the primary benchmark.

The synthetic benchmark demonstrates practical strengths. Bootstrap temporal entropy detects 74.1% of flatline events versus 24.1% for conventional temporal measurements. For independent noise, conventional recall is 1.000 versus 0.463 for temporal entropy. These responses occur at unequal achieved background alarm rates. Windows spanning a flatline onset can retain measurable entropy, while wholly constant windows trigger abstention. Paired results distinguish mechanism, severity, duration, measured direction and affected data availability. Direction denotes the observed lagged difference. The first decision after onset occurs 11 samples later, highlighting how window length and decision frequency influence detection of short events.

6.3 Reference quality and practical measurement requirements

For RQ5, Table 2 demonstrates the value of checking whether generated trajectories reproduce the statistics used for alarms. The paired design distinguishes model fidelity, estimator availability and alarm calibration as separate factors in monitoring performance. Coverage measures how often observations fall within generated 90% intervals. Synthetic64 spatial entropy coverage is 0.453 for diffusion versus 0.740 for bootstrap. Intel values are 0.083 versus 1.000, averaging six shared spatial comparisons per issuance. Intel diffusion nevertheless improves the raw energy score (6.336 versus 7.234), while sample entropy coverage is 0.000 versus 0.953. This contrast provides a concrete criterion for generative monitoring. A reference must represent the dynamics underlying the detector's features as well as predict sensor values.

RQ6 identifies how to make more measurements usable (Fig. 5). On Intel, moving from S to T increases eligible group/window fractions from 0.017 to 0.541 under bootstrap and from 0.049 to 0.633 under diffusion. Temporal estimators use individual sensor histories without requiring complete rows across an entire group. Calibration also depends on this availability. Restricting Intel bootstrap B to common feature support and recalibrating changes background exceedance from 0.000 to 0.551. Thus, practical evaluation must check achieved alarm rates under changing data availability and calibration/test distributions.

Development selects $q = 3$, with delay 1 for synthetic data and 2 for both real datasets. Patterns with $q = 4$ lack sufficient support in these windows. Tolerance and group fraction are selected per configuration. At $W = 96$, scale 2 can retain 30 templates, while scale 4

Table 1. Fault detection and sensor localization at nominal $\alpha = .10$. Block denotes joint bootstrap. P and R denote event precision and recall. AP is event precision-recall envelope area. IoU measures overlap between identified and affected sensors, assigning zero to missed events. Background alarm rates differ across methods.

Synthetic (three sizes)					
Reference	Family	P	R	AP	IoU
Block	S	0.256	0.285	0.220	0.024
Block	T	0.329	0.401	0.290	0.093
Block	ST	0.308	0.409	0.272	0.075
Block	B	0.484	0.650	0.452	0.193
Block	BST	0.486	0.653	0.456	0.200
Diffusion	S	0.197	0.170	0.196	0.010
Diffusion	T	0.227	0.318	0.209	0.044
Diffusion	ST	0.197	0.170	0.196	0.009
Diffusion	B	0.327	0.392	0.284	0.102
Diffusion	BST	0.327	0.392	0.284	0.103

Intel Lab					
Reference	Family	P	R	AP	IoU
Block	S	0.251	0.208	0.083	0.017
Block	T	0.329	0.389	0.172	0.076
Block	ST	0.329	0.389	0.172	0.070
Block	B	0.000	0.000	0.139	0.000
Block	BST	0.000	0.000	0.139	0.000
Diffusion	S	0.158	0.213	0.086	0.014
Diffusion	T	0.048	0.060	0.051	0.002
Diffusion	ST	0.048	0.060	0.051	0.002
Diffusion	B	0.444	0.056	0.120	0.015
Diffusion	BST	0.444	0.056	0.120	0.011

PEMS-BAY					
Reference	Family	P	R	AP	IoU
Block	S	0.211	0.528	0.179	0.035
Block	T	0.291	0.389	0.229	0.015
Block	ST	0.291	0.389	0.229	0.018
Block	B	0.224	0.148	0.207	0.030
Block	BST	0.224	0.148	0.207	0.030
Diffusion	S	0.373	0.287	0.281	0.006
Diffusion	T	0.276	0.403	0.225	0.011
Diffusion	ST	0.376	0.287	0.300	0.001
Diffusion	B	0.267	0.111	0.211	0.022
Diffusion	BST	0.267	0.111	0.211	0.022

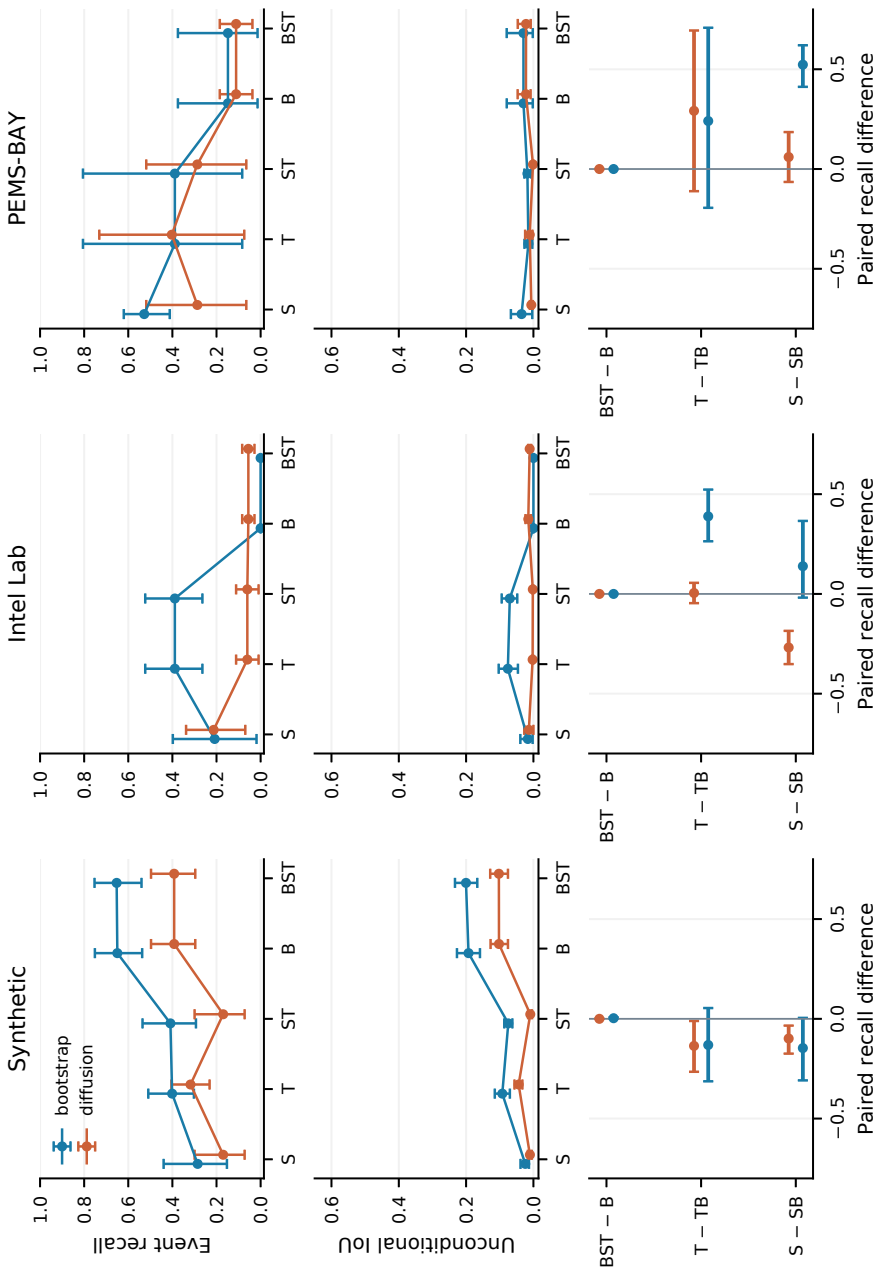


Fig. 3. Detection, localization and unadjusted paired 95% intervals obtained by resampling whole blocks. Missed faults receive zero IoU. Equal nominal α permits different background alarm rates. Four real recording blocks per dataset limit statistical precision.

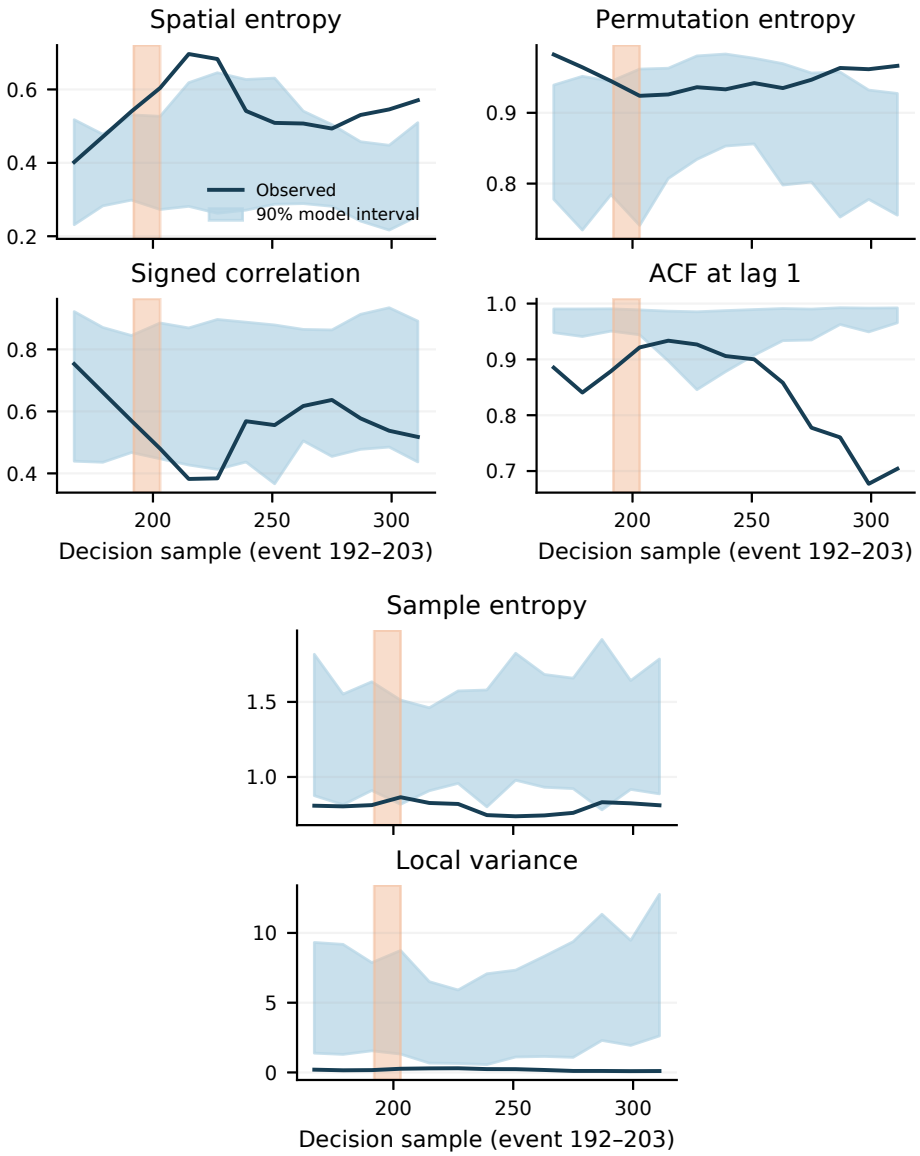


Fig. 4. First prespecified synthetic copying event. Shading marks the 12 injected observations. All forecast bands use the same generated trajectories. Injection labels select the first affected sensor and best-overlap candidate group for illustration only, independently of the detection and localization rules.

Table 2. Reference quality on unchanged backgrounds using seed 17, the first two blocks and decisions 167/263. Evaluated observations or features are shared across references, although donor gaps affect estimator support. Raw, H, PE and SE report coverage of 90% intervals. TV measures ordinal-pattern total variation. ES is normalized energy score, with lower values better. Temporal entries use $W = 96$. Synthetic rows combine three sizes. Undefined estimates are omitted. Counts and interval widths are reported separately.

Dataset	Reference	Raw	H	PE	SE
Synthetic	Block	0.875	0.812	0.855	0.858
Synthetic	Diffusion	0.803	0.513	0.429	0.222
Intel Lab	Block	0.616	1.000	0.887	0.953
Intel Lab	Diffusion	0.714	0.083	0.214	0.000
PEMS-BAY	Block	0.847	0.819	0.873	0.862
PEMS-BAY	Diffusion	0.739	0.494	0.699	0.430

Dataset	Reference	ACF1 MAE	Ordinal TV	R RMSE	ES
Synthetic	Block	0.034	0.081	0.236	0.578
Synthetic	Diffusion	0.047	0.150	0.313	1.747
Intel Lab	Block	0.013	0.378	0.467	7.234
Intel Lab	Diffusion	0.114	0.406	0.525	6.336
PEMS-BAY	Block	0.121	0.094	0.288	1.501
PEMS-BAY	Diffusion	0.163	0.118	0.360	2.055

cannot. $W = 24$ also lacks support for the primary temporal entropies. These diagnostics identify feasible window and scale combinations.

Requiring 52 rather than 51 valid spatial reference draws leaves recall unchanged (0.000 difference) in all five seed configurations after recalibration. Separate $W = 48/96$ comparisons use window-specific thresholds while retaining the 120-step forecast lead, keeping the prediction task fixed.

Exploratory scale2 tests reuse seed 17 episodes, $W = 96$ histories, $d = 24$ and both references. Temporal measurements average pairs of observations. Bootstrap T recall changes from scale 1 to 2 are -0.093 $[-0.231, +0.069]$, -0.069 $[-0.458, +0.333]$ and -0.333 $[-0.722, -0.028]$ for synthetic, Intel and PEMS. Corresponding TB changes are -0.074 $[-0.245, +0.042]$, $+0.264$ $[+0.000, +0.750]$ and -0.153 $[-0.278, -0.028]$. Each comparison includes localization, availability and achieved background rates. Spatial S/SB scores remain unchanged, and scale-1 calibration reproduces the independent $W = 96$ check. These comparisons characterize scale sensitivity without selecting a new primary scale.

A subsequent CPU study of 128 independent Gaussian AR(1) simulations with unit variance provides guidance on estimator stability. For independent continuous observations, increasing W from 48 to 96 reduces PE standard deviation from 0.023 to 0.009. At sample-entropy tolerance .2, finite supported fractions rise from 0.086 to 1.000. Longer windows therefore improve both stability and availability in this setting. At $\rho = .8$, rounding to .5-unit steps produces tied-template fraction 0.653. PE eligibility is 0.008 with deterministic tie ordering versus 0.758 after discarding tied templates.

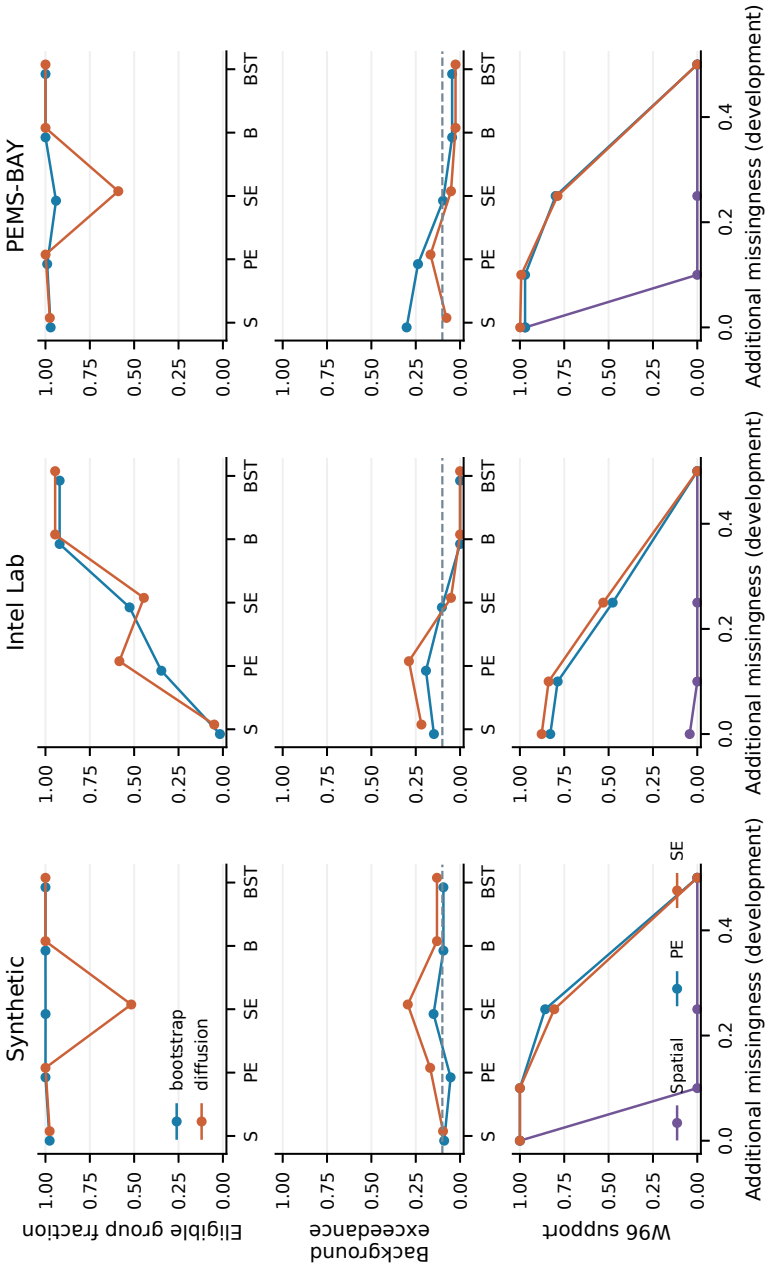


Fig. 5. Top panels show the fraction of eligible groups. Middle panels show background alarm rates at nominal $\alpha = .10$. Bottom panels show availability after independently adding missingness during development, using $W = 96$. Spatial availability is measured per group, while PE and SE availability is measured per sensor.

This quantifies how sensor resolution and tie handling affect measurement availability, supporting explicit quality checks before interpreting entropy.

7 Dashboard and GPU Computation

The dashboard links alerts to sensor behavior. Operators can inspect contributing sensors, coordinated changes and departures from generated references. Fixed sensor positions support visual comparison, with separate overlays for physical connections and correlation changes.

Selecting a group reveals individual entropy traces, conventional measurements and predictive bands. The display separates actual increases or decreases from departures relative to expected behavior. Template and pair counts indicate whether enough observations support an estimate. Shared channels, change direction and onset timing help users investigate an alert without assigning a physical cause. Forecast and observation times, model version and the scan calibrated p-value provide context. Three prespecified copying examples demonstrate the workflow without selecting for success.

CUDA batches correlation matrices, eigenvalues, ordinal patterns, conventional statistics and joint trajectory generation. Sample entropy runs in bounded batches to limit temporary memory. Statistical calculations use float32 and are checked against CPU float64, separately from neural mixed precision. Stored predictions, calibration results, missingness information and reference trajectories support reconstruction of the displayed evidence.

On an RTX A6000, median time from reference generation through sensor localization is 0.180, 0.174 and 0.258 seconds for bootstrap on synthetic64, Intel and PEMS, respectively. Diffusion takes 0.370, 0.338 and 1.164 seconds. Peak allocated memory is at most 1.43 GiB. These timings include missingness handling, both windows, feature extraction, scoring and localization. They exclude GDN, data acquisition, loading, disk output, rank lookup and dashboard rendering.

For PEMS, the temporal calculation takes 0.724 seconds on CPU float64 and 0.035 seconds on GPU float32, excluding transfers. The comparison combines hardware and precision differences, while maximum numerical disagreement remains below 2.0×10^{-7}. Timed GPU stages totaled 4.271 hours across experiments, validation and interrupted attempts. The core computation is shorter than the real datasets' sampling intervals. Overall response time also includes data handling, window accumulation and decision scheduling.

8 Conclusion

This study provides an integrated approach to evaluating the entropy for IoT fault detections. The contribution is the joint assessment of measurement information, data availability, generative reference quality and alarm calibration, supported by GPU computation and an inspectable dashboard. Spatial and temporal measurements capture different changes, supporting feature selection according to sensor behavior and fault mechanisms.

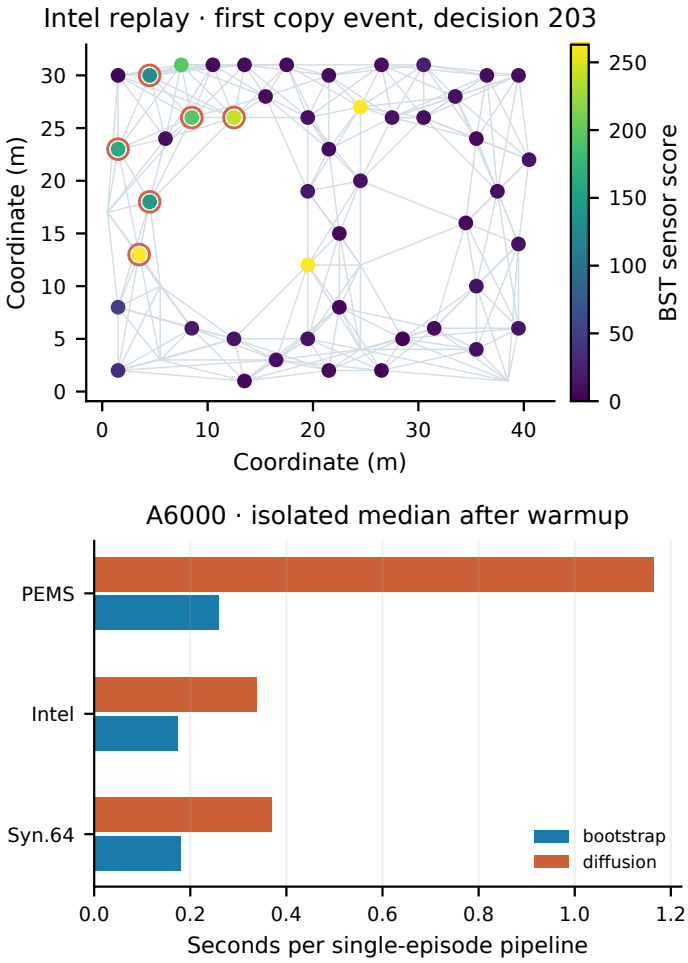


Fig. 6. Inspecting sensor groups and computational cost. The Intel view replays the first prespecified copying trial. Orange outlines mark injected sensors for evaluation only. Timing covers reference generation through localization and excludes data acquisition, loading and dashboard presentation.

Temporal entropy expands availability on incomplete Intel data and improves detection under the evaluated bootstrap calibration, with unequal background rates. Conventional temporal measurements provide stronger synthetic localization. Adding entropy increases synthetic bootstrap recall by 0.003 and leaves other primary recalls unchanged, identifying limited incremental benefit under the tested maximum rule. Neither entropy nor the evaluated diffusion reference shows a consistent overall advantage. A central finding is that better raw predictive scores need not produce faithful entropy distributions. This gives generative monitoring a concrete evaluation requirement and helps distinguish model deficiencies from measurement limitations.

The evidence concerns injected faults and partly reused recordings, with four blocks per real dataset. Forecast lead, calibration size, estimator support and candidate groups shape the conclusions. The framework supports validation on native faults and longer matched histories, together with operator studies toward practical deployment.

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